Hidden in the Leftovers

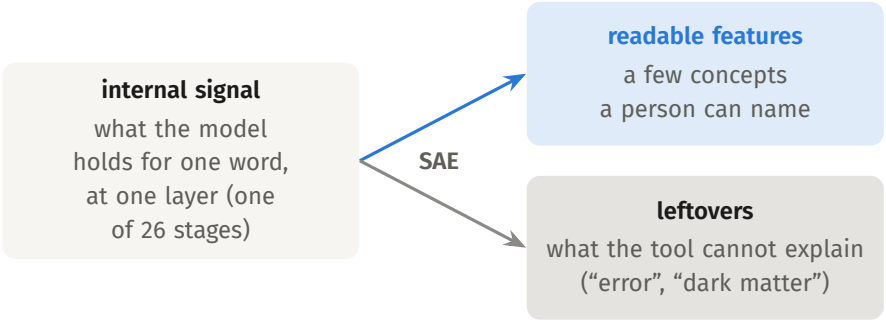
Task-specific blind spots of SAEs, in English and Indian languages

[Student name]

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[Course name]

Why · We read language models with sparse autoencoders



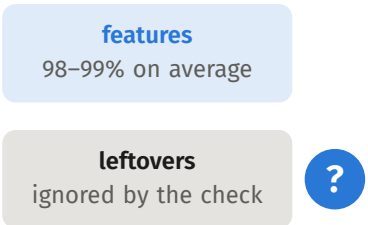
Sparse autoencoder (SAE): a leading tool to read a model’s internal signal; used to explain, audit and steer models [1, 6].

Standard check: next-word prediction with the SAE swapped in stays **98–99%** as good.

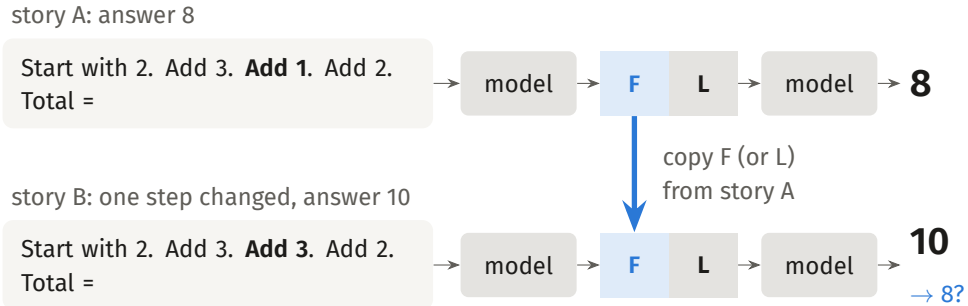
Why · The worry: the real work may hide in the leftovers

- What SAEs miss, their “dark matter” [3], may hold the computation that matters.
- Then explanations built from features miss the important part.
- The standard check averages over ordinary text: **it cannot tell**.

Pilot question: when a model keeps track of something, is it in the features or the leftovers?



Pilot · The swap test: does the right answer come back?



F = readable features, **L** = leftovers, copied at one layer. **Score:** share of the answer brought back, 0% (still 10) to 100% (back to 8). Copying everything: 100%.

Pilot · Four tasks the model can solve

Running sum

`Start with 2. Add 3. Add 1. Add 2. Total =` **8**

keeps a total

Object tracking

`The key is in box B. … The key moves to box C. … The key is in box` **C**

links objects to boxes

Variable copying

`t = 7. q = 4. x = q. q = 2. x =` **4**

follows an assignment

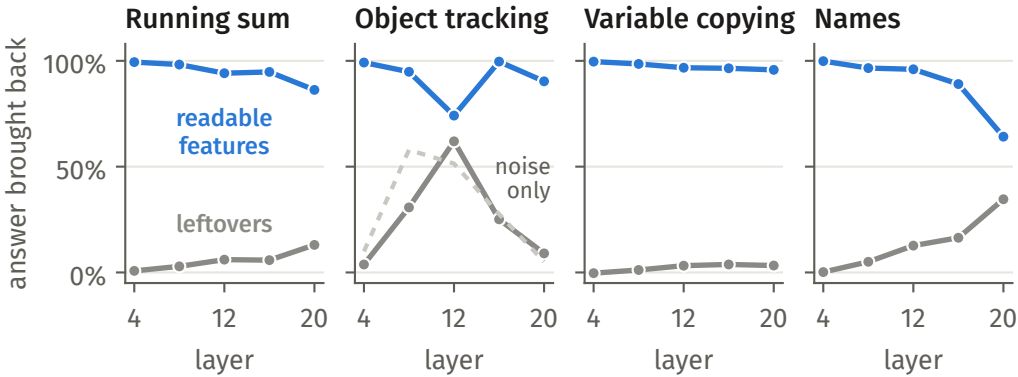
Names (control)

`When Mary and John went to the store, John gave a drink to` **Mary**

a well-understood case [8]

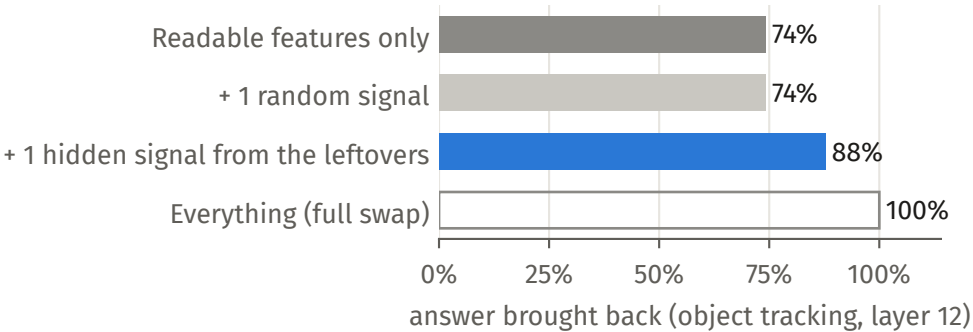
gemma-2-2b (Google's open 2-billion-parameter model) · Gemma Scope SAEs [7]
150 test pairs per task · the 3 tasks first planned were too hard for the model

Pilot · Finding 1: on tracking tasks the features carry 74–100%



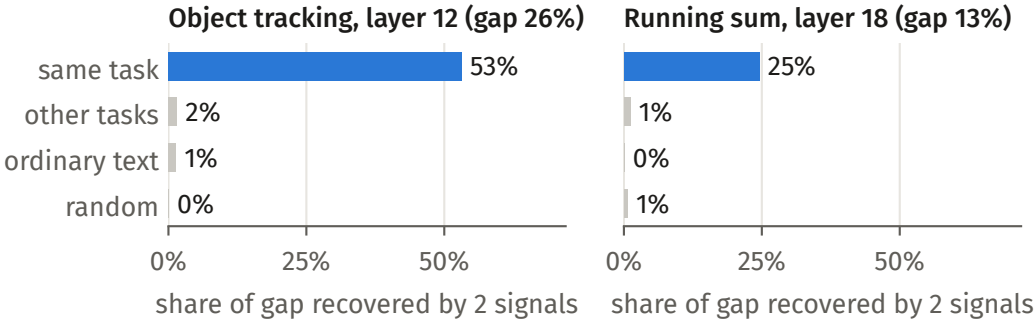
Leftovers alone: little. Object tracking's grey bump: mostly a noise effect (dashed).

Pilot · Finding 2: passes the standard check, yet misses 26% here



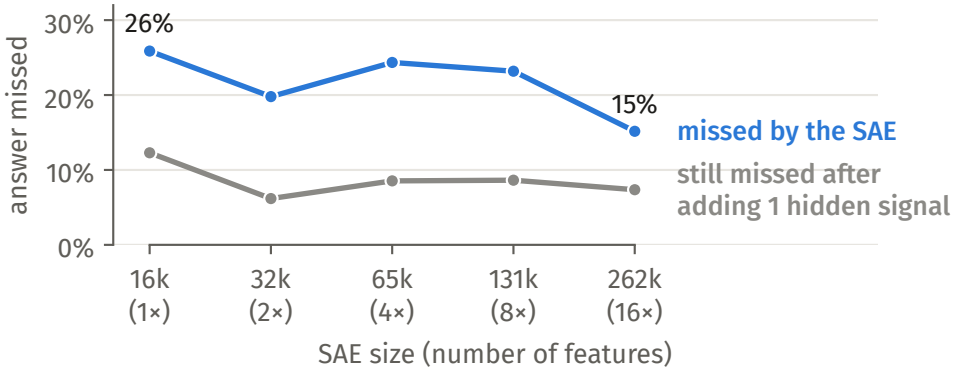
Hidden signal: top pattern in the leftovers, learned on 300 pairs, tested on 150 new ones.

Pilot · Finding 3: the hidden signal belongs to the task



Same task: a real share. Other tasks or ordinary text: almost nothing.

Pilot · Finding 4: a 16× bigger SAE shrinks the gap, but 15% remains



One hidden signal still closes 51–69% of what is left.

Caveat: bigger SAEs also switch on more features per word; Q1 separates the two.

What · One blind spot found. How many more?

SAEs can have blind spots that averages cannot see.

Q1

Which tasks have them?

more tasks; how big; how specific

Q2

What is the hidden signal?

binding: how the model links an object to its box?

Q3

Bigger in Indian languages?

Hindi and Hinglish first

How · Q1 and Q2: which tasks, and is the signal binding?

Kind of task	Examples	Status
Linking things	objects in boxes; people and their cities	pilot + new
Keeping a count	running sum; longer chains	pilot + new
Following references	variable copying	pilot
Control	names	pilot

Q1: per task: how big is the gap, is it task-specific, does size or sparsity fix it?
Success: gap $\geq$ 10%, and same-task signals close $\geq$ 30 points more than any other source.

Q2: binding signals exist in other models [4]. Edit only the hidden signal: other box named?
Success: $\geq$ 50% of pairs switch box, against $\leq$ 5% for a random signal.

How · Q3: are the blind spots bigger in Indian languages?

English (pilot baseline)

Hindi

Hinglish (Latin script)

Tamil or Bengali (if time)

French (non-Indian control)

SAEs trained on mostly English text [7]. Prior work: language-specific features [2], not what is lost.

Why: SAE audits are used for Hindi speakers too; if SAEs miss more there, those audits mean less.

Three checks per language

1. Standard check on Hindi text vs English
2. Swap test: is the gap bigger?
3. Does the English hidden signal work in Hindi?

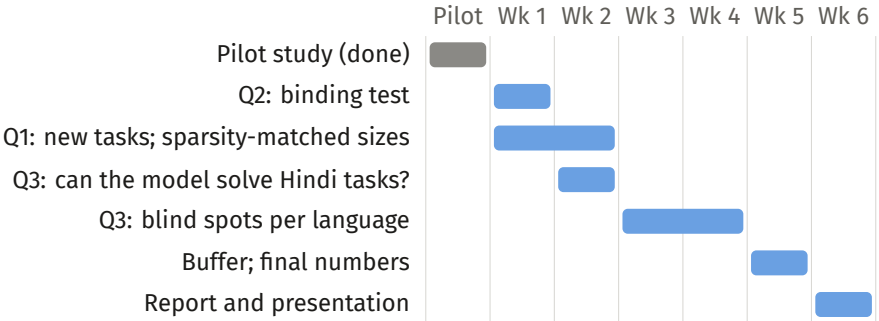
Success: larger Hindi gap, non-overlapping 95% intervals.

Translated tasks (draft, to be checked)

शुरुआत 2 से करें। 3 जोड़ें। 1 जोड़ें। 2 जोड़ें। कुल = **8**

Rahul aur Priya market gaye; Priya ne book di. Book kisko mili? **Rahul**

When · Six weeks, three deliverables



Deliverables: **blind-spot map** · **swap-test script** · **binding verdict**

Feasible: the pilot ran on one 8 GB laptop GPU in about 6 h.

Hidden in the leftovers

SAEs look near-perfect on average, but can hide task-specific blind spots.

- **Found:** object tracking misses 26%; one hidden signal from that task brings back half.
- **Proposed:** map blind spots across tasks and in Hindi and Hinglish; test if the signal is binding.
- **Takeaway:** before trusting an SAE explanation, run the swap test, task by task.

References

[1] T. Bricken et al. Towards monosemanticity: Decomposing language models with dictionary learning. Transformer Circuits Thread, 2023. Published October 4, 2023.

[2] B. Deng et al. Unveiling language-specific features in large language models via sparse autoencoders. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 4563–4608, July 2025. Association for Computational Linguistics.

[3] J. Engels, L. R. Smith, and M. Tegmark. Decomposing the dark matter of sparse autoencoders. *Transactions on Machine Learning Research*, 2025.

[4] J. Feng and J. Steinhardt. How do language models bind entities in context? In *The Twelfth International Conference on Learning Representations (ICLR)*, 2024.

[5] J. Hewitt and P. Liang. Designing and interpreting probes with control tasks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 2733–2743, Nov. 2019. Association for Computational Linguistics.

[6] R. Huben et al. Sparse autoencoders find highly interpretable features in language models. In *The Twelfth International Conference on Learning Representations (ICLR)*, 2024.

[7] T. Lieberum et al. Gemma Scope: Open sparse autoencoders everywhere all at once on Gemma 2. In *Proceedings of the 7th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*, pages 278–300, Nov. 2024. Association for Computational Linguistics.

[8] K. R. Wang et al. Interpretability in the wild: a circuit for indirect object identification in GPT-2 small. In *The Eleventh International Conference on Learning Representations (ICLR)*, 2023.

Thank you

Questions?

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Backup slides

B1 · Setup and how we measure

Setup

- gemma-2-2b, 26 layers, 2304 numbers per word
- Gemma Scope SAEs, 16k to 262k features
- 300 training / 150 test pairs per task; no prompt shared
- 95% bootstrap confidence intervals
- Code: 78 unit tests; matches Hugging Face (same top answer on 95% of prompts)

Measures

- Score = share of the answer brought back (0 to 1)
- Gap = 1 − score of copying the features only
- Hidden signal: top pattern of the leftovers on training pairs
- Controls: random signals; leftovers replaced by noise
- Standard check: 16k SAEs, layers 10–18, ordinary text (the Pile)

B2 · Three pitfalls that would each have misled us

Copy the whole sentence

47% vs 100%

object tracking, layer 12: last word only vs from the first changed word

Rule out word shortcuts

100%

a reader of the words alone (no model) solves variable copying and names [5]

Readable ≠ used

84% vs 47%

box read from the leftovers (chance 18%), yet copying that word brings back 47%

B3 · The original claims and what the pilot found

Original claim	Pilot verdict
The leftovers carry what the model tracks	**Rejected:** on the three tracking tasks, features bring back 74–100%; leftovers 0–13% (object tracking up to 62%, mostly a noise effect)
The state sits in the leftovers in a simple shape	**Readable, not used:** object tracking's box is read from the leftovers (84%), but the model does not use it at the last word
A few signals in the leftovers fill the gap	**Yes, narrowly:** one signal closes half of object tracking's 26% gap at layer 12, and it is specific to that task

B4 · Caveats

- One model, one SAE family, synthetic tasks; size tested at layer 12 only.
- Two kinds of gap: names loses 24–36% at layers 18–20, spread over about 32 signals (layer 18).
- Object tracking's "leftovers only" 62% is inflated: leftovers replaced by noise already give 51%.
- Bigger SAEs use more features per word (124–139 vs 78–89 at 16k): size and sparsity are confounded.
- Our pre-set rule (widest SAE keeps $\geq$ 60% of the gap) is narrowly missed: 15.2% vs 15.5%.
- Binding and everything multilingual are plans, not results.